Operating Systems · 21CSC202J

Unit 1 · Session 3 · SLO 2 — Solved Worksheet

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Unit â€“ 1 : Session â€“ 3 : SLO â€“ 2 | Multiprogramming / Time-Sharing Word Search + Kernel Structures

I. Word Search â€” Words to Find

• MONITOR, MULTITASKING, PROCESSOR, VIEW, ROUND ROBIN, SHORTEST, SIMULTANEOUSLY, TIME SHARING
• All eight words are located in the grid (horizontal, vertical, and diagonal directions).
• Hidden message from the unused letters: "MULTIPROGRAMMING IMPROVES CPU UTILIZATION".

II. Kernel Structures â€” Reverse Word Search (place these words)

• EXOKERNEL â€” minimalist kernel that exposes hardware to applications, giving them direct control.
• HYBRID KERNEL â€” combines features of monolithic and microkernels; used in Windows and macOS.
• LAYERED â€” organized in levels; each layer builds on services provided by the one below it.
• MICROKERNEL â€” only essential functions like IPC and basic scheduling; everything else in user space.
• MODULAR â€” components can be loaded and unloaded at runtime; used in Linux systems (LKMs).
• MONOLITHIC â€” single large process running entirely in kernel mode with all OS services included.
• SIMPLE â€” basic structure with minimal functionality, used in teaching or early systems (e.g. MS-DOS).
• VIRTUAL MACHINE (HYPERVISOR) â€” abstracts physical hardware; runs multiple OSes on the same system.